6.5 From analysis to architecture

Analysis object (from 5.1 and 5.2)	Component that owns it (6.3)	Note (split, merged, external)
StudentAccount	Authentication Component	Manages student accounts, login information and access to the system
InternshipApplication	Application Management Component	Handles creating, viewing, editing, and deleting internship applications
Status Change	Status Tracking Component	Updates application statuses and stores the history of status change
Interview	Application Management Component	Stores interview dates and other important details related to an application
Document	Document Management Component	Handles PDF uploads and storage
Reminder	Reminder Management Component	Handles upcoming deadlines and interviews
ProgressSummary	Dashboard	Displays application progress and counts

6.6 Design principles applied

Principle	Where it shows in your design	Evidence (component, interface, ADR)
Separation of concerns	Layered architecture in Section 6.2 and component diagram in 6.3	Layered architecture(ADR-01), web user interface, data access
Information hiding	Authentication and Document Management Components	Account and file storage details are managed within their components
Low coupling	Component interfaces in section 6.3	Application Management uses the AccessCheck and ApplicationStore interfaces instead of directly handling authentication and database access
High cohesion	Application Management, Status tracking and Reminder Components	Each component focuses on related tasks
Design for change	Layered architecture and separate document storage	Layered architecture(ADR-01) and separate document storage(ADR-02) allow components to be updated independently.

Principle	Where it shows in your design	Evidence (component, interface, ADR)
Keep it simple (KISS)	Layered architecture selected in Section 6.2	Avoids unnecessary microservices and complex integrations

7 Design Rationale

7.1 Architecture decision records

ADR-01 [Layered Architecture]

Status	Accepted
Date and owner	11/07/2026 , Oluwanifemi Olayiwola
Context (drivers)	InternTrack needs to manage student accounts, internship applications, reminders, and uploaded documents. The system should be organized, secure and easy to maintain as new features are added
Options considered	Layered architecture, two-tier client server and microservices
Decision	We plan to use a layered architecture because it keeps the system organized and makes it easier to develop, test, and update different parts of InternTrack
Consequences	The system is easier to build, test and update. The requests may take longer because they pass through different layers. If the application server fails, the system may stop working
Evidence	Architecture comparison in 6.2 and diagrams in 6.3- 6.4
Related	QAS-01, QAS-02, QAS-03

ADR-02 [Separate Document Storage]

Status	Accepted
Date and owner	11/07/2026 , Oluwanifemi Olayiwola
Context (drivers)	FR-12, QR-02, document storage requirements
Options considered	• Separate File Storage: stores the actual documents separately and saves their information in the database • Database Storage: stores the documents directly in the database with the other application information
Decision	We plan to use separate file storage because it keeps documents organized and makes them easier to manage.
Consequences	This makes the system easier to maintain and update document storage. However, we need to make sure the documents stay connected to the correct student and application. The application can become a

Evidence	Context diagram in Section 2.1 and document storage design in sections 6.3-6.4
Related	FR-12, QR-02, QAS-02,, CH-01

ADR-03 [In-App reminders]

Status	Accepted
Date and owner	11/07/2026 , Oluwanifemi Olayiwola
Context (drivers)	FR-09 requires reminders for upcoming interviews and deadlines within seven days. CON-03 excludes email and text notifications.
Options considered	• In-App Reminders: Shows reminders inside InternTrack when students use the website • External Notifications: Sends reminders through email or text messages.
Decision	We plan to use in-app reminders because they are simpler to build and do not need outside services.
Consequences	This keeps the system simple and makes it easy to maintain. But students will need to open the InternTrack to see their reminders.
Evidence	FR-09 and CON-03 in section 3, UC-03 in section 3.4 and reminder management in section 6.3
Related	FR-09, CON-03, UC-03

7.2 Trade-off summary

Quality favored	Architectural move	Cost accepted	Recorded in
Maintainability and security	Use layered architecture to separate the interface, application rules, domain logic, and data storage	Requests must pass through multiple layers, which may increase processing time	ADR-01
Maintainability and flexibility	Keep uploaded documents in separate file storage and hide storage details behind the Document Management component.	Additional file management and permission checks	ADR-02
Simplicity	Use dashboard reminders instead of email or text notifications	Students must open the website in order to see reminders.	ADR-03